

BG77xA-GL&BG95xA-GL

SSL Application Note

LPWA Module Series

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About the Document

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1 Introduction

Quectel BG77xA-GL and BG95xA-GL modules support SSL function.

SSL (Secure Sockets Layer) is a networking protocol designed for securing connections between web clients and web servers over an insecure network, such as the internet.

The SSL function is to ensure the privacy of communication. In some cases, the communication between the server and the client should be encrypted to prevent tampering, eavesdropping attacks, or data forging.

1.1. Applicable Modules

Table 1: Applicable Modules

Module Series	Model
BG77xA-GL	BG770A-GL
	BG773A-GL
BG95xA-GL	BG950A-GL
	BG951A-GL
	BG953A-GL
	BG955A-GL

1.2. SSL Versions and Cipher Suites

The following SSL versions are supported by BG770A-GL, BG950A-GL and BG951A-GL.

Table 2: Supported SSL Versions

SSL Version
SSL3.0
TLS1.0
TLS1.1
TLS1.2

SSL cipher suites supported by the modules are presented in the table below. For a detailed description of cipher suites, see **document [1]**.

Table 3: Supported SSL Cipher Suites

Cipher Suite Code	Cipher Suite Name
0X0035	TLS_RSA_WITH_AES_256_CBC_SHA
0X002F	TLS_RSA_WITH_AES_128_CBC_SHA
0X0005	TLS_RSA_WITH_RC4_128_SHA
0X0004	TLS_RSA_WITH_RC4_128_MD5
0X000A	TLS_RSA_WITH_3DES_EDE_CBC_SHA
0X003D	TLS_RSA_WITH_AES_256_CBC_SHA256
0XC002	TLS_ECDH_ECDSA_WITH_RC4_128_SHA
0XC003	TLS_ECDH_ECDSA_WITH_3DES_EDE_CBC_SHA
0XC004	TLS_ECDH_ECDSA_WITH_AES_128_CBC_SHA
0XC005	TLS_ECDH_ECDSA_WITH_AES_256_CBC_SHA
0XC007	TLS_ECDHE_ECDSA_WITH_RC4_128_SHA
0XC008	TLS_ECDHE_ECDSA_WITH_3DES_EDE_CBC_SHA
0XC009	TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA
0XC00A	TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA
0XC011	TLS_ECDHE_RSA_WITH_RC4_128_SHA

0XC012	TLS_ECDHE_RSA_WITH_3DES_EDE_CBC_SHA
0XC013	TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA
0XC014	TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA
0XC00C	TLS_ECDH_RSA_WITH_RC4_128_SHA
0XC00D	TLS_ECDH_RSA_WITH_3DES_EDE_CBC_SHA
0XC00E	TLS_ECDH_RSA_WITH_AES_128_CBC_SHA
0XC00F	TLS_ECDH_RSA_WITH_AES_256_CBC_SHA
0XC023	TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256
0XC024	TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA384
0XC025	TLS_ECDH_ECDSA_WITH_AES_128_CBC_SHA256
0XC026	TLS_ECDH_ECDSA_WITH_AES_256_CBC_SHA384
0XC027	TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256
0XC028	TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA384
0XC029	TLS_ECDH_RSA_WITH_AES_128_CBC_SHA256
0XC02A	TLS_ECDH_RSA_WITH_AES_256_CBC_SHA384
0XC02B	TLS-ECDHE-ECDSA-WITH-AES-128-GCM-SHA256
0XC02F	TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256
0XC0A8	TLS_PSK_WITH_AES_128_CCM_8
0X00AE	TLS_PSK_WITH_AES_128_CBC_SHA256
0XFFFF	Support all cipher suites above

1.3. Using SSL Function

Step 1: Configure <APN>, <username>, <password> and other parameters of a PDP context with **AT+QICSGP**. See **document [2]** for detailed information.

Step 2: Activate the PDP context with **AT+QIACT**, then the assigned IP address can be queried with **AT+QIACT?**. See **document [2]** for detailed information.

Step 3: Configure the SSL version, cipher suite, trusted CA certificate path and the security level for a specified SSL context with **AT+QSSLCFG**.

Step 4: Open an SSL socket to connect to a remote server with **AT+QSSLOPEN**. **<SSL_ctxID>** is used to specify SSL context, and **<access_mode>** is used to specify data access mode.

Step 5: After the SSL connection has been established, data will be sent or received via the connection. For detailed information about how to send and receive data in each data access mode, see **Chapter 1.4**.

Step 6: Close SSL connection with **AT+QSSLCLOSE**.

Step 7: Deactivate the PDP context with **AT+QIDEACT**. See **document [2]** for detailed information.

1.4. Description of Data Access Modes

The SSL connection supports three data access modes:

- Buffer access mode
- Direct push mode
- Transparent transmission mode

When opening an SSL connection via **AT+QSSLOPEN**, you can specify the data access mode with **<access_mode>**. After the SSL connection is established, the data access mode can be switched with **AT+QISWTMD** (see **document [2]**).

1. In buffer access mode, the module buffers data upon receiving them and reports a URC in the format of **+QSSLURC: "recv",<clientID>** to notify the host of the incoming data. In this case, the host can retrieve the buffered data with **AT+QSSLRECV**.
2. In direct push mode, the module outputs the received data directly through a URC in the format of **+QSSLURC: "recv",<clientID>,<current_recv_length><CR><LF><data>**.
3. In transparent transmission mode, the corresponding COM port is exclusively used for sending/receiving data directly to/from the Internet. It cannot be used for other purposes such as running AT commands, etc.

- **Exit transparent transmission mode**

To make the module exit transparent transmission mode either:

- 1) Execute **+++**. Follow the requirements below to prevent the **+++** from being misinterpreted as data:
 - a) Do not input any character for at least 1 second before and after inputting **+++**.

- b) Input **+++** within 1 second, and wait until **OK** is returned. After **OK** is returned, the module switches to buffer access mode.

OR

- 2) Change MAIN_DTR from LOW to HIGH to make the module enter command mode (the COM port can now be used for running AT commands, as well as for sending/retrieving data). In this case, set **AT&D1** (see *document [3]*) before the module enters transparent transmission mode.

- **Return to transparent transmission mode**

To return to transparent transmission mode either:

- 1) Execute **AT+QISWTMD**. Before execution specify **<access_mode>** as 2. Once transparent transmission mode is entered successfully, **CONNECT** is returned.

OR

- 2) Execute **ATO**. After a connection exits transparent transmission mode, executing **ATO** switches the data access mode back to transparent transmission mode. Once transparent transmission mode is entered successfully, **CONNECT** is returned. If no connection has entered transparent transmission mode, **ATO** returns **NO CARRIER**.

NOTE

1. In buffer access mode, if the buffer is not empty and the module receives data again, it does not report a new URC until all the received data have been retrieved with **AT+QSSLRECV** from the buffer.
2. In transparent transmission mode, AT commands cannot be executed. If the socket connection is closed because of a network error or other errors, the module reports **NO CARRIER** and exits the transparent transmission mode. In this case, execute **AT+QSSLCLOSE** to close the socket service.

1.5. Certificate Validity Check

To check certificate validity, compare your local time against certificate validity range ("Not before" and "Not after"). If the local time is earlier than the "Not before" time or later than the "Not after" time the certificate has expired.

When certificate validity check is required (set **<ignore_localetime>** as 0 when executing **AT+QSSLCFG**), to avoid validity period check failure, execute **AT+CCLK** to configure the module time within the certificate validity period. See *document [3]* for detailed information about **AT+CCLK**.

1.6. Server Name Indication

SNI (Server Name Indication) allows the server to safely host multiple TLS Certificates since it provides Server Host Name information as an extension in the client hello message. It thus enhances connection security with multiple virtual servers based on a single IP address. This feature is only applicable to TLS protocol.

2 Description of SSL AT Commands

2.1. AT Command Introduction

2.1.1. Definitions

- **<CR>** Carriage return character.
- **<LF>** Line feed character.
- **<...>** Parameter name. Angle brackets do not appear on the command line.
- **[...]** Optional parameter of a command or an optional part of TA information response. Square brackets do not appear on the command line. When an optional parameter is not given in a command, the new value equals its previous value or the default settings, unless otherwise specified.
- **Underline** Default setting of a parameter.

2.1.2. AT Command Syntax

All command lines must start with **AT** or **at** and end with **<CR>**. Information responses and result codes always start and end with a carriage return character and a line feed character: **<CR><LF><response><CR><LF>**. In tables presenting commands and responses throughout this document, only the commands and responses are presented, and **<CR>** and **<LF>** are deliberately omitted.

Table 4: Types of AT Commands

Command Type	Syntax	Description
Test Command	AT+<cmd>=?	Test the existence of the corresponding command and return information about the type, value, or range of its parameter.
Read Command	AT+<cmd>?	Check the current parameter value of the corresponding command.
Write Command	AT+<cmd>=<p1>[,<p2>[,<p3>[...]]]	Set user-definable parameter value.
Execution Command	AT+<cmd>	Return a specific information parameter or perform a specific action.

2.2. Declaration of AT Command Examples

The AT command examples in this document are provided to help you learn about the use of the AT commands introduced herein. The examples, however, should not be taken as Quectel's recommendations or suggestions about how to design a program flow or what status to set the module into. Sometimes multiple examples may be provided for one AT command. However, this does not mean that there is a correlation among these examples, or that they should be executed in a given sequence.

2.3. Description of SSL AT Commands

2.3.1. AT+QSSLCFG Configure Parameters of an SSL Context

This command configures the SSL version, cipher suite, security level, CA certificate, client certificate and client key. These parameters will be used in the handshake procedure.

<SSL_ctxID> is the index of the SSL context. The module supports 6 SSL contexts at most. Several SSL connections can be established based on one SSL context. The settings such as the SSL version and the cipher suite are stored in the SSL context, and they will be applied to the new SSL connections associated with the SSL context.

AT+QSSLCFG Configure Parameters of an SSL Context	
Test Command	Response
AT+QSSLCFG=?	+QSSLCFG: "sslversion",(range of supported <SSL_ctxID>s),(range of supported <SSL_version>s) +QSSLCFG: "ciphersuite",(range of supported <SSL_ctxID>s),(list of supported <cipher_suites>s) +QSSLCFG: "cacert",(range of supported <SSL_ctxID>s),<cacertpath> +QSSLCFG: "clientcert",(range of supported <SSL_ctxID>s),<clientcertpath> +QSSLCFG: "clientkey",(range of supported <SSL_ctxID>s),<clientkeypath> +QSSLCFG: "seclevel",(range of supported <SSL_ctxID>s),(range of supported <seclevel>s) +QSSLCFG: "session",(range of supported <SSL_ctxID>s),(list of supported <session>s) +QSSLCFG: "sni",(range of supported <SSL_ctxID>s),(list of supported <SNI>s) +QSSLCFG: "checkhost",(range of supported <SSL_ctxID>s),(list of supported <check_host>s) +QSSLCFG: "ignorelocaltime",(range of supported

	<SSL_ctxID>s),(list of supported <ignore_localtime>s) +QSSLCFG: "negotiatetime",(range of supported <SSL_ctxID>s),(range of supported <negotiate_time>s) +QSSLCFG: "renegotiation",(range of supported <SSL_ctxID>s),(list of supported <renegotiation>s) +QSSLCFG: "dtls",(range of supported <SSL_ctxID>s),(list of supported <DTLS_enable>s) +QSSLCFG: "dtlsversion",(range of supported <SSL_ctxID>s),(range of supported <DTLS_version>s) OK
Write Command Query/configure the SSL version for a specified SSL context: AT+QSSLCFG="sslversion",<SSL_ctxID>[,<SSL_version>]	Response If the optional parameter is omitted, query the current setting. +QSSLCFG: "sslversion",<SSL_ctxID>,<SSL_version> OK If the optional parameter is specified, set the SSL version for the specified SSL context. OK Or ERROR
Write Command Query/configure the SSL cipher suites for a specified SSL context: AT+QSSLCFG="ciphersuite",<SSL_ctxID>[,<cipher_suites>]	Response If the optional parameter is omitted, query the current setting. +QSSLCFG: "ciphersuite",<SSL_ctxID>,<cipher_suites> OK If the optional parameter is specified, set the SSL cipher suite for the specified SSL context. OK Or ERROR
Write Command Query/configure the path of trusted CA certificate for a specified SSL context: AT+QSSLCFG="cacert",<SSL_ctxID>[,<cacertpath>]	Response If the optional parameter is omitted, query the current setting. +QSSLCFG: "cacert",<SSL_ctxID>,<cacertpath> OK

	If the optional parameter is specified, set the path of trusted CA certificate for the specified SSL context. OK Or ERROR
Write Command Query/configure the path of client certificate for a specified SSL context: AT+QSSLCFG="clientcert",<SSL_ctxID>[,<clientcertpath>]	Response If the optional parameter is omitted, query the current setting. +QSSLCFG: "clientcert",<SSL_ctxID>,<clientcertpath> OK If the optional parameter is specified, set the path of client certificate for the specified SSL context. OK Or ERROR
Write Command Query/configure the path of client private key for a specified SSL context: AT+QSSLCFG="clientkey",<SSL_ctxID>[,<clientkeypath>]	Response If the optional parameter is omitted, query the current setting. +QSSLCFG: "clientkey",<SSL_ctxID>,<clientkeypath> OK If the optional parameter is specified, set the path of client private key for the specified SSL context. OK Or ERROR
Write Command Query/configure the authentication mode for a specified SSL context: AT+QSSLCFG="seclevel",<SSL_ctxID>[,<seclevel>]	Response If the optional parameter is omitted, query the current setting. +QSSLCFG: "seclevel",<SSL_ctxID>,<seclevel> OK If the optional parameter is specified, set the authentication mode for the specified SSL context. OK Or ERROR

Write Command Query/configure SSL Resumption feature for a specified SSL context: AT+QSSLCFG="session",<SSL_ctxID>[,<session>]	Response If the optional parameter is omitted, query the current setting. +QSSLCFG: "session",<SSL_ctxID>,<session> OK If the optional parameter is specified, enable/disable SSL Resumption feature for the specified SSL context. OK Or ERROR
Write Command Query/configure Server Name Indication feature for a specified SSL context: AT+QSSLCFG="sni",<SSL_ctxID>[,<SNI>]	Response If the optional parameter is omitted, query the current setting. +QSSLCFG: "sni",<SSL_ctxID>,<SNI> OK If the optional parameter is specified, enable/disable Server Name Indication feature for the specified SSL context. OK Or ERROR
Write Command Query/configure hostname validation feature for a specified SSL context: AT+QSSLCFG="checkhost",<SSL_ctxID>[,<check_host>]	Response If the optional parameter is omitted, query the current setting. +QSSLCFG: "checkhost",<SSL_ctxID>,<check_host> OK If the optional parameter is specified, enable/disable hostname validation feature for the specified SSL context. OK Or ERROR
Write Command Query/configure whether to ignore certificate validity check for a specified SSL context: AT+QSSLCFG="ignorelocaltime",<SSL_ctxID>[,<ignore_localtime>]	Response If the optional parameter is omitted, query the current setting. +QSSLCFG: "ignorelocaltime",<SSL_ctxID>,<ignore_localtime>

	OK If the optional parameter is specified, set whether or not to ignore certificate validity check for the specified SSL context. OK Or ERROR
Write Command Query/configure the maximum timeout of SSL negotiation for a specified SSL context: AT+QSSLCFG="negotiatetime",<SSL_ctxID>[,<negotiate_time>]	Response If the optional parameter is omitted, query the current setting. +QSSLCFG: "negotiatetime",<SSL_ctxID>,<negotiate_time> OK If the optional parameter is specified, set the maximum timeout of SSL negotiation for the specified SSL context. OK Or ERROR
Write Command Query/configure whether support for TLS renegotiation is enabled for a specified SSL context: AT+QSSLCFG="renegotiation",<SSL_ctxID>[,<renegotiation>]	Response If the optional parameter is omitted, query the current setting. +QSSLCFG: "renegotiation",<SSL_ctxID>,<renegotiation> OK If the optional parameter is specified, set whether to enable support for TLS renegotiation for the specified SSL context. OK Or ERROR
Write Command Query the current setting or configure whether to enable DTLS feature for a specified SSL context: AT+QSSLCFG="dtls",<SSL_ctxID>[,<DTLS_enable>]	Response If the optional parameter is omitted, query the current setting. +QSSLCFG: "dtls",<SSL_ctxID>,<DTLS_enable> OK If the optional parameter is specified, set whether to

	enable DTLS feature for the specified SSL context. OK Or ERROR
Write Command Query/configure the DTLS version of a specified SSL context: AT+QSSLCFG="dtlsversion",<SSL_ctxID>[,<DTLS_version>]	Response If the optional parameter is omitted, query the current setting. +QSSLCFG: "dtlsversion",<SSL_ctxID>,<DTLS_version> OK If the optional parameter is specified, set the DTLS version of the specified SSL context. OK Or ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configurations will not be saved.

Parameter

<SSL_ctxID>	Integer type. SSL context ID. Range: 0–5.
<SSL_version>	Integer type. SSL Version. 0 SSL3.0 1 TLS1.0 2 TLS1.1 3 TLS1.2 4 All
<cipher_suites>	Numeric type in HEX format. SSL cipher suites. 0X0035 TLS_RSA_WITH_AES_256_CBC_SHA 0X002F TLS_RSA_WITH_AES_128_CBC_SHA 0X0005 TLS_RSA_WITH_RC4_128_SHA 0X0004 TLS_RSA_WITH_RC4_128_MD5 0X000A TLS_RSA_WITH_3DES_EDE_CBC_SHA 0X003D TLS_RSA_WITH_AES_256_CBC_SHA256 0XC002 TLS_ECDH_ECDSA_WITH_RC4_128_SHA 0XC003 TLS_ECDH_ECDSA_WITH_3DES_EDE_CBC_SHA 0XC004 TLS_ECDH_ECDSA_WITH_AES_128_CBC_SHA 0XC005 TLS_ECDH_ECDSA_WITH_AES_256_CBC_SHA 0XC007 TLS_ECDHE_ECDSA_WITH_RC4_128_SHA 0XC008 TLS_ECDHE_ECDSA_WITH_3DES_EDE_CBC_SHA

	0XC009	TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA
	0XC00A	TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA
	0XC011	TLS_ECDHE_RSA_WITH_RC4_128_SHA
	0XC012	TLS_ECDHE_RSA_WITH_3DES_EDE_CBC_SHA
	0XC013	TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA
	0XC014	TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA
	0XC00C	TLS_ECDH_RSA_WITH_RC4_128_SHA
	0XC00D	TLS_ECDH_RSA_WITH_3DES_EDE_CBC_SHA
	0XC00E	TLS_ECDH_RSA_WITH_AES_128_CBC_SHA
	0XC00F	TLS_ECDH_RSA_WITH_AES_256_CBC_SHA
	0XC023	TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256
	0XC024	TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA384
	0XC025	TLS_ECDH_ECDSA_WITH_AES_128_CBC_SHA256
	0XC026	TLS_ECDH_ECDSA_WITH_AES_256_CBC_SHA384
	0XC027	TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256
	0XC028	TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA384
	0XC029	TLS_ECDH_RSA_WITH_AES_128_CBC_SHA256
	0XC02A	TLS_ECDH_RSA_WITH_AES_256_CBC_SHA384
	0XC02B	TLS-ECDHE-ECDSA-WITH-AES-128-GCM-SHA256
	0XC02F	TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256
	0XC0A8	TLS_PSK_WITH_AES_128_CCM_8
	0X00AE	TLS_PSK_WITH_AES_128_CBC_SHA256
	0XFFFF	Support all cipher suites
<cacertpath>	String type. The path of the trusted CA certificate.	
<clientcertpath>	String type. The path of the client certificate.	
<clientkeypath>	String type. The path of the client private key.	
<secclevel>	Integer type. Authentication mode.	
	0	No authentication
	1	Perform server authentication
	2	Perform server and client authentication if requested by the remote server
<session>	Integer type. Enable or disable SSL Resumption feature.	
	0	Disable
	1	Enable
<SNI>	Integer type. Enable or disable Server Name Indication feature. DNS hostnames are currently the only supported server names.	
	0	Disable
	1	Enable
<check_host>	Integer type. Enable or disable hostname validation feature (Subject Common Name (CN) matches the specified host name).	
	0	Disable
	1	Enable
<ignore_localtime>	Integer type. Whether or not to ignore certificate validity check.	
	0	Not to ignore
	1	Ignore

<negotiate_time>	Integer type. The maximum timeout of SSL negotiation. Range: 10–300. Default value: 300. Unit: second.
<renegotiation>	Integer type. Enable or disable support for TLS renegotiation. 0 Disable 1 Enable
<DTLS_enable>	Integer type. Enable or disable DTLS feature. 0 Disable 1 Enable
<DTLS_version>	Integer type. DTLS version. 0 DTLS1.0 1 DTLS1.2 2 Both

2.3.2. AT+QSSLOPEN Open an SSL Socket to Connect to a Remote Server

This command sets up an SSL connection. During the negotiation between the module and the Internet, parameters configured with **AT+QSSLCFG** will be used in the handshake procedure. After successful handshake with the Internet, the module can send or receive data via this SSL connection. In addition, the module can set up several SSL connections based on one SSL context.

According to steps mentioned in **Chapter 1.3**, execute **AT+QIACT** first to activate the PDP context and then execute **AT+QSSLOPEN**. It is suggested to wait for a specific period of time (refer to the Maximum Response Time below) for **+QSSLOPEN: <clientID>,<result>** URC to be outputted. If the URC is not received during that time, use **AT+QSSLCLOSE** to close the SSL connection.

AT+QSSLOPEN Open an SSL Socket to Connect to a Remote Server	
Test Command AT+QSSLOPEN=?	Response +QSSLOPEN: (range of supported <PDP_ctxID>s),(range of supported <SSL_ctxID>s),(range of supported <clientID>s),<serveraddr>,<server_port>,(range of supported <access_mode>s) OK
Write Command AT+QSSLOPEN=<PDP_ctxID>,<SSL_ctxID>,<clientID>,<serveraddr>,<server_port>[,<access_mode>]	Response If the <access_mode>=2 (transparent transmission mode) and the SSL connection is successfully set up: CONNECT If the <access_mode>=0/1(buffer access mode or direct push mode): OK +QSSLOPEN: <clientID>,<result>

	If there is any error: ERROR
Maximum Response Time	Maximum network response time of 150 s, plus configured time of <negotiate_time> .
Characteristics	The command takes effect immediately. The configurations will not be saved.

Parameter

<PDP_ctxID>	Integer type. PDP context ID. Range: 1–5.
<SSL_ctxID>	Integer type. SSL context ID. Range: 0–5.
<clientID>	Integer type. Socket index. Range: 0–11.
<serveraddr>	String type. Remote server address.
<server_port>	Integer type. Listening port of remote server. Range: 0-65535.
<access_mode>	Integer type. SSL connection data access mode. 0 Buffer access mode 1 Direct push mode 2 Transparent transmission mode
<result>	Integer type. Result code. See Chapter 5 .
<negotiate_time>	Integer type. Maximum timeout of SSL negotiation. Range: 10–300. Default value: 300. Unit: second.

NOTE

Error description can be got via **AT+QIGETERROR**. For more details, refer to **document [2]**.

2.3.3. AT+QSSSEND Send Data via SSL Connection

After the connection is established, the module can send data through the SSL connection.

AT+QSSSEND Send Data via SSL Connection	
Test Command AT+QSSSEND=?	Response +QSSSEND: (range of supported <clientID> s),(range of supported <sendlen> s) OK
Write Command Send variable-length data AT+QSSSEND=<clientID>	Response > After the above response, input the data to be sent. Tap CTRL+Z to send, or tap ESC to cancel the operation.

	If the connection has been established and sending is successful: SEND OK If the connection has been established but sending buffer is full: SEND FAIL If the connection has not been established, abnormally closed, or any parameter is incorrect: ERROR
Write Command Send fixed-length data AT+QSSSEND=<clientID>,<sendlen>	Response > After the above response, input the data until the data length equals <sendlen>. If the connection has been established and sending is successful: SEND OK If the connection has been established but sending buffer is full: SEND FAIL If the connection has not been established, abnormally closed, or the parameter is incorrect: ERROR
Maximum Response Time	300 ms
Characteristics	/

Parameter

<clientID>	Integer type. Socket index. Range: 0–11.
<sendlen>	Integer type. Length of data to be sent. Range: 1–1460. Unit: byte.

2.3.4. AT+QSSLRCV Retrieve Data via SSL Connection

When an SSL connection is opened with <access_mode> specified as 0 (buffer access mode), the module reports URC **+QSSLURC: "recv",<clientID>** when it receives data from the Internet. You can read the data from buffer with **AT+QSSLRCV**.

AT+QSSLRCV Retrieve Data via SSL Connection	
Test Command AT+QSSLRCV=?	Response +QSSLRCV: (range of supported <clientID>s),(range of supported <readlen>s) OK
Write Command When <readlen> is not 0, retrieve the data in specified length AT+QSSLRCV=<clientID>[,<readlen>]	Response If the specified connection has received data: +QSSLRCV: <read_actual_length><CR><LF><data> OK When the buffer is empty: +QSSLRCV: 0 OK If connection cannot be established, abnormally closed, or the parameter is incorrect: ERROR
Write Command When <readlen> is 0, query the retrieved data length AT+QSSLRCV=<clientID>,0	Response If the specified connection exists: +QSSLRCV: <total_receive_length>,<have_read_length>,<unread_length> OK If there is any error: ERROR
Maximum Response Time	300 ms
Characteristics	/

Parameter

<clientID>	Integer type. Socket index. Range: 0–11.
<readlen>	Integer type. Length of data to be retrieved. Range: 1–1500. Default value: 1500. Unit: byte.
<read_actual_length>	Integer type. Length of the data that have been actually retrieved. Unit: byte.
<data>	String type. Retrieved data.
<total_receive_length>	Integer type. Total length of received data. Unit: byte.
<have_read_length>	Integer type. The actual data length read with AT+QSSLRCV . Unit: byte.

<unread_length>	Integer type. Length of the data that have not been retrieved. Unit: byte.
-----------------	--

2.3.5. AT+QSSLCLOSE Close an SSL Connection

This command closes an SSL connection. If all the SSL connections based on the same SSL context are closed, the module releases the SSL context.

AT+QSSLCLOSE Close an SSL Connection	
Test Command AT+QSSLCLOSE=?	Response +QSSLCLOSE: (range of supported <clientID>s),(range of supported <close_timeout>s) OK
Write Command AT+QSSLCLOSE=<clientID>[,<close_timeout>]	Response If the SSL connection is successfully closed: OK If there is any error: ERROR
Maximum Response Time	10 s by default, determined by parameter <close_timeout>
Characteristics	The command takes effect immediately. The configuration will not be saved.

Parameter

<clientID>	Integer type. Socket index. Range: 0–11.
<close_timeout>	Integer type. Timeout value of AT+QSSLCLOSE . Range: 0–65535. Default value: 10. Unit: second. 0 means closing immediately.

2.3.6. AT+QSSLSTATE Query the State of SSL Connections

This command queries the state of SSL connections.

AT+QSSLSTATE Query the State of SSL Connections	
Test Command AT+QSSLSTATE=?	Response OK
Write Command AT+QSSLSTATE=<clientID>	Response +QSSLSTATE: <clientID>,"SSLClient",<IP_address>,<remote_port>,<local_port>,<socket_state>,<PDP_ctxID>,<serverID>,<access_mode>,<AT_port>,<SSL_ctxID>

	OK
Execution Command AT+QSSLSTATE	Response List of (+QSSLSTATE: <clientID>,"SSLClient",<IP_address>,<remote_port>,<local_port>,<socket_state>,<PDP_ctxID>,<serverID>,<access_mode>,<AT_port>,<SSL_ctxID>)s
	OK
Maximum Response Time	300 ms
Characteristics	/

Parameter

<clientID>	Integer type. Socket index. Range: 0–11.
<IP_address>	String type. Remote server address.
<remote_port>	Integer type. Remote server port. Range: 0–65535.
<local_port>	Integer type. Local port. Range: 0–65535.
<socket_state>	Integer type. SSL connection state. 0 "Initial" Connection has not been established 1 "Opening" Client is connecting 2 "Connected" Connection has been established 4 "Closing" Connection is closing
<PDP_ctxID>	Integer type. PDP context ID. Range: 1–16.
<serverID>	Integer type. Reserved.
<access_mode>	Integer type. SSL connection data access mode. 0 Buffer access mode 1 Direct push mode 2 Transparent transmission mode
<AT_port>	String type. COM port of socket service. "main" Main UART port "aux" AUX UART port on BG770A-GL and Debug UART on BG95xA-GL "emux0" EMUX0 "emux1" EMUX1 "emux2" EMUX2 "emux3" EMUX3
<SSL_ctxID>	Integer type. SSL context ID. Range: 0–5.

2.3.7. AT+QSSLCRYPT Encrypt/Decrypt Data with a Specified Algorithm

This command encrypts/decrypts data with a specified algorithm. It supports AES-128-CBC encryption/decryption and MD5 encryption.

AT+ QSSLCRYPT Encrypt/Decrypt Data with a Specified Algorithm	
Test Command AT+QSSLCRYPT=?	Response +QSSLCRYPT: ("AES","MD5"),<input_data>,(list of supported <format>s),(list of supported <mode>s),<key> OK
Write Command AT+QSSLCRYPT="AES",<input_data>,<format><mode>,<key>	Response +QSSLCRYPT: "AES",<output_data> OK If there is any error: ERROR
Write Command AT+QSSLCRYPT="MD5",<input_data>	Response +QSSLCRYPT: "MD5",<output_data> OK If there is any error: ERROR
Maximum Response Time	300 ms
Characteristic	/

Parameter

<input_data>	In AES-128-CBC mode: The data to be encrypted or decrypted. The data format is specified by <format> . The maximum length of <input_data> is 256 bytes. The length of data to be encrypted/decrypted with AES-128-CBC should be a multiple of 16 bytes. In MD5 mode: String type. The data to be encrypted. The maximum length of <input_data> is 256 bytes.
<format>	Integer type. The format of input data. 0 String format 1 Hex format
<mode>	Integer type. AES-128-CBC mode. 0 AES-128-CBC decryption 1 AES-128-CBC encryption
<key>	String type. The encryption/decryption key of AES-128-CBC. The length must be 16 bytes.
<output_data>	Hex String type. The encrypted data or decrypted data.

2.4. Description of URCs

2.4.1. +QSSLURC: "recv" Notify Received Data

The URC notifies the data received from peer in buffer access mode and direct push mode.

+QSSLURC: "recv" Notify Received Data	
+QSSLURC: "recv",<clientID>	The URC of SSL data incoming in buffer access mode. SSL data can be retrieved with AT+QSSLRECV .
+QSSLURC: "recv",<clientID>,<current_recv_length><CR><LF><data>	The URC of SSL data incoming in direct push mode.

Parameter

<clientID>	Integer type. Socket index. Range: 0–11.
<current_recv_length>	Integer type. Length of actual received data. Unit: byte.
<data>	The actual received data.

2.4.2. +QSSLURC: "closed" Notify Abnormal Disconnection

The URC notifies that the SSL connection has been disconnected. Disconnection can be caused by many reasons. For example, the Internet closes the connection or the state of GPRS PDP is deactivated, and the SSL connection state based on the specified socket may be "closing". In such case, **AT+QSSLCLOSE=<connectID>** must be executed to change the SSL connection state to "initial".

+QSSLURC: "closed" Notify Abnormal Disconnection	
+QSSLURC: "closed",<clientID>	The SSL connection based on the specified socket is closed.

Parameter

<clientID>	Integer type. Socket index. Range: 0–11.
------------	--

3 Examples

3.1. Configure and Activate a PDP Context

3.1.1. Configure a PDP Context

```
AT+QICSGP=1,1,"CMNBIOT","","",1 //Configure PDP context as 1. China Mobile NB-IoT network
OK                                APN: CMNBIOT.
```

3.1.2. Activate a PDP Context

```
AT+QIACT=1 //Activate PDP context 1.
OK          //Activated successfully.
AT+QIACT?   //Query the context state, protocol type and IP address of
+QIACT: 1,1,1,"100.142.162.0" PDP context 1.
OK
```

3.1.3. Deactivate a PDP Context

```
AT+QIDEACT=1 //Deactivate PDP context 1.
OK           //Deactivated successfully.
```

3.2. Configure an SSL Context

```
AT+QSSLCFG="sslversion",1,1 //Set SSL context ID as 1 and SSL version as TLS1.0.
OK
AT+QSSLCFG="ciphersuite",1,0X0035 //Set SSL cipher suites of SSL context ID 1 as
OK                                TLS_RSA_WITH_AES_256_CBC_SHA.
AT+QSSLCFG="secllevel",1,1 //Set authentication mode of SSL context ID 1 as server
OK                                authentication.
```

```
OK
AT+QSSLCFG="cacert",1,"cacert.pem" //Set path of the trusted CA certificate of SSL context ID 1
OK
```

3.3. SSL Client in Buffer Access Mode

3.3.1. Set up an SSL Connection and Enter Buffer Access Mode

```
AT+QSSLOPEN=1,1,0,"220.180.239.212",8010,0 //Set up an SSL connection.
OK

+QSSLOPEN: 0,0 //SSL connection has been set up successfully.
AT+QSSLSTATE //Query the state of all SSL connections.
+QSSLSTATE: 0,"SSLClient","220.180.239.212",8010,65344,2,1,4,0,"main",1
OK
```

3.3.2. Send Data in Buffer Access Mode

```
AT+QSSLSEND=0 //Send variable-length data.
>
Test data from SSL
<CTRL+Z>
SEND OK
AT+QSSLSEND=0,18 //Send fixed-length data with the data length of 18 bytes.
>
Test data from SSL
SEND OK
```

3.3.3. Retrieve Data in Buffer Access Mode

```
+QSSLURC: "recv",0 //The Socket 0 (<clientID>=0) has received data.

AT+QSSLRCV=0,1500 //Retrieved data. Length of data to be retrieved:1500 bytes.
+QSSLRCV: 18 //Length of retrieved data: 18 bytes.
Test data from SSL

OK
AT+QSSLRCV=0,1500
+QSSLRCV: 0 //The buffer is empty.
```

OK

3.3.4. Close an SSL Connection

```
AT+QSSLCLOSE=0 //Close the connection (<clientID>=0). Depending on the
network, the maximum response time is 10 s by default.
OK
```

3.4. SSL Client in Direct Push Mode

3.4.1. Set up an SSL Connection and Enter Direct Push Mode

```
AT+QSSLOPEN=1,1,0,"220.180.239.212",8011,1 //Set up an SSL connection.
OK

+QSSLOPEN: 0,0 //SSL connection has been set up successfully.
AT+QSSLSTATE //Query the state of all SSL connections.
+QSSLSTATE: 0,"SSLClient","220.180.239.212",8011,65047,2,1,4,1,"main",1
OK
```

3.4.2. Send Data in Direct Push Mode

```
AT+QSSLSEND=0 //Send variable-length data.
>
Test data from SSL
<CTRL+Z>
SEND OK
AT+QSSLSEND=0,18 //Send fixed-length data. Data length is 18 bytes.
>
Test data from SSL
SEND OK
```

3.4.3. Retrieve Data in Direct Push Mode

```
+QSSLURC: "recv",0,18
Test data from SSL
```


3.4.4. Close an SSL Connection

```
AT+QSSLCLOSE=0 //Close the connection (<clientID>=0). Depending on the
                network, the maximum response time is 10 s by default.

OK
```

3.5. SSL Client in Transparent Transmission Mode

3.5.1. Set up an SSL Connection and Send Data in Transparent Transmission Mode

```
AT+QSSLOPEN=1,1,0,"220.180.239.212",8011,2 //Set up an SSL connection.
CONNECT //Enter transparent transmission mode.
        //The client is sending data from COM port to the Internet directly.
        (The data are not visible in the example.)

OK //Use +++ or MAIN_DTR (set AT&D1 first) to exit transparent
   transmission mode. The NO CARRIER result code indicates that the
   server has stopped the SSL connection.
```

3.5.2. Set up an SSL Connection and Retrieve Data in Transparent Transmission Mode

```
AT+QSSLOPEN=1,1,0,"220.180.239.212",8011,2 //Set up an SSL connection.
CONNECT //Enter transparent transmission mode.
<Received data> //The client is reading the data.

OK //Use +++ or MAIN_DTR (execute AT&D1 first) to exit transparent
   transmission mode. The NO CARRIER result code indicates that the
   server has stopped the SSL connection.
```

3.5.3. Close an SSL Connection

```
AT+QSSLCLOSE=0 //Close the connection (<clientID>=0). Depending on the network, the
                maximum response time is 10 s by default.

OK
```

3.6. DTLS Connection Based on PSK Encryption

3.6.1. Set up a DTLS Connection

```
AT+QFUPL="0_server.psk" //Upload PSK file first. The PSK file should be named as
                        SSL_ctxID_server.psk (0_server.psk for instance) and the
```

content of the file should be in the format of
<PSK_ID>&<PSK_key> (e.g., DTLS_Client&1a2b3c4d).

CONNECT

<Input file bin data>

+QFUPL: 24,553

OK

AT+QSSLCFG="dtls",0,1

//Enable DTLS feature for SSL context 0.

OK

AT+QSSLCFG="dtlsversion",0,0

//Configure DTLS version to DTLS1.0 for SSL context 0.

OK

AT+QSSLCFG="ciphersuite",0,0X00AE

//Configure cipher suite for SSL context 0.

OK

AT+QSSLOPEN=1,0,0,"220.180.239.201",8010,0

OK

+QSSLOPEN: 0,0

//SSL connection has been set up successfully.

AT+QSSLSTATE

//Query the state of SSL connections.

+QSSLSTATE: 0,"SSLClient","220.180.239.201",8010,65344,2,1,4,0,"main",1

OK

3.6.2. Close an SSL Connection

AT+QSSLCLOSE=0

//Close the connection (<clientID>=0). Depending on the network, the maximum response time is 10 s by default.

OK

4 Check for Failure in SSL Connection

To identify reasons for the failure to open an SSL connection:

1. Query the status of the specified PDP context with **AT+QIACT?** to check whether the specified PDP context has been activated.
2. If the address of server is a domain name, check if the DNS server address is valid with **AT+QIDNSCFG=<contextID>**. Because an invalid DNS server address cannot convert domain name to IP address.
3. Check the SSL configuration with **AT+QSSLCFG**, especially the SSL version and cipher suite, to make sure that they are supported on server side. If **<seclevel>** has been configured as 1 or 2, then the trusted CA certificate has to be uploaded to the module with **AT+QFUPL**. If the server has configured "SSLVerifyClient required", the client certificate and client private key have to be uploaded to the module with **AT+QFUPL**. For detailed information about certificate validity check, see **Chapter 1.5**. And for more information about related FILE AT commands, see **document [4]**.

5 Summary of Result Codes

The result code **<result>** indicates an error related to mobile equipment or network. The meaning of **<result>** is described in the following table.

Table 5: Result Codes

<result>	Meaning
0	Operation successful
550	Unknown error
551	Operation blocked
552	Invalid parameter
553	Memory not enough
554	Create socket failed
555	Operation not supported
556	Socket bind failed
557	Socket listen failed
558	Socket write failed
559	Socket read failed
560	Socket accept failed
561	Open PDP context failed
562	Close PDP context failed
563	Socket identity has been used
564	DNS busy
565	DNS parse failed

566	Socket connection failed
567	Socket has been closed
568	Operation busy
569	Operation timeout
570	PDP context break down
571	Cancel send
572	Operation not allowed
573	APN not configured
574	Port busy

6 Appendix References

Table 6: Related Documents

Document Name
[1] RFC 2246 – The TLS Protocol Version 1.0
[2] Quectel_BG77xA-GL&BG95xA-GL_TCP(IP)_Application_Note
[3] Quectel_BG77xA-GL&BG95xA-GL_AT_Commands_Manual
[4] Quectel_BG77xA-GL&BG95xA-GL_FILE_Application_Note

Table 7: Terms and Abbreviations

Abbreviation	Description
APN	Access Point Name
CA	Certificate Authority
CR	Carriage Return
DNS	Domain Name Server
DTR	Data Terminal Ready
DTLS	Datagram Transport Layer Security
GPRS	General Packet Radio Service
ID	Identifier
IP	Internet Protocol
LF	Line Feed
PDP	Packet Data Protocol
SNI	Server Name Indication

SSL	Secure Sockets Layer
TCP/IP	Transmission Control Protocol/Internet Protocol
TLS	Transport Layer Security
UART	Universal Asynchronous Receiver/Transmitter
URC	Unsolicited Result Code
USB	Universal Serial Bus